

$$SG = \frac{s}{s_w} = \frac{8}{1000}$$

$$s_{kr} = 800$$

$$s_{Lox} = 1140$$

$$P_T = \dot{m}_P \left[\frac{\frac{\Delta P_o}{s}}{\eta_P} \right]$$

$$\eta_P = 65\%$$

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$$P_{ox}$$

$$\Delta P_o = 4.6 \text{ MPa}$$

$$P_f$$

$$\Delta P_o = 3.8 \text{ MPa}$$

$$\dot{m} = 92 \text{ kg/s}$$

$$r = 2.5$$

$$\frac{\dot{m}_{ox}}{\dot{m}_{fuel}} = 2.5$$

T.C

$$P_p = P_{p_{ox}} + P_{p_{fuel}}$$

$$= \dot{m}_{ox} \left[\frac{\frac{\Delta P_{o_{ox}}}{s_{ox}}}{\eta_{p_{ox}}} \right] + \dot{m}_{fuel} \left[\frac{\frac{\Delta P_{o_{fuel}}}{s_{fuel}}}{\eta_{p_{fuel}}} \right]$$

$$\frac{\dot{m}_{ox}}{\dot{m}_{fuel}} = 2.5 \Rightarrow \begin{cases} \dot{m}_{ox} + \dot{m}_{fuel} = 92 \text{ kg/s} \\ 3.5 \dot{m}_{fuel} = 92 \text{ kg/s} \end{cases}$$

$$\dot{m}_{fuel} = 26.3 \text{ kg/s}$$

$$\dot{m}_{ox} = 92 - 26.3 = 65.7 \text{ kg/s}$$